

QUANTUM RANDOM NUMBER GENERATOR (QRNG)

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Technologies: Python, IBM Qiskit, Matplotlib

1. Business Problem

Random number generation is essential for scientific simulations, statistical sampling, gaming, and digital security. Biased or poorly evaluated random-number generators can reduce the reliability of software systems.

Our project explores quantum measurement-based random-bit generation and compares its statistical performance with Python's classical pseudorandom-number generator.

2. Target Customers and Business Value

Potential users: Simulation developers, gaming companies, research institutions, and quantum computing educators.

Business value: A low-cost, reproducible prototype for studying quantum circuits and evaluating random-bit distributions against a classical baseline.

Illustrative cost: If investigating a randomness-related software defect takes six engineering hours at ₹600/hour, the investigation would cost ₹3,600. This is a hypothetical scenario, not an observed customer expense or demonstrated saving.

3. Proposed Solution

We developed a QRNG prototype using Qiskit.

- Initialize one qubit in state $|0\rangle$.
- Apply a Hadamard gate to establish equal measurement probabilities.
- Measure the circuit over 4,096 simulation shots.
- Generate 4,096 classical pseudorandom bits using Python.
- Analyze bit balance, chi-square statistics, and p-values.
- Create comparative distribution and bit-sequence graphs.

4. Experimental Results

Parameter Qiskit Simulator Python Random

Total bits	4,096	4,096
Zeros	2,043	2,108
Ones	2,053	1,988
Ones (%)	50.12%	48.54%
Chi-square	0.0244	3.5156
P-value	0.8758	0.0608

Key finding: Both generators produced approximately balanced distributions and showed no statistically significant deviation from 50/50 bit balance at the 5% significance level. The quantum simulator produced a more balanced sample in this run, but this does not prove it is a superior random-number generator.

5. Limitations and Future Scope

The current implementation uses Qiskit's BasicSimulator rather than physical IBM quantum hardware. It therefore does not demonstrate physical quantum entropy, cryptographic security, or quantum advantage.

Bit-balance and chi-square tests provide limited statistical evidence; additional testing is necessary.

Future improvements: IBM quantum hardware execution, larger sample testing, additional randomness tests, and hardware noise analysis.

6. Conclusion

This project demonstrates quantum circuit fundamentals through an operational QRNG prototype, supported by classical benchmarking, statistical evaluation, and data visualization. It offers a reproducible educational foundation for exploring quantum randomness and its practical applications.